resisto-GEM: strain-specific GEM reconstruction pipeline

v0.2 (as implemented) · 2026-10-22 · Nextflow · AMRBench_QC gene prediction, CarveMe 1.6.6, then species-reference biomass and gap-filling

PER GENOME · 140 NOW, 12,000 LATER (7 SPECIES)

Assemblies

`data/genomes/*/*.fna`
samplesheet:
`sample, species, fasta`
species selects the reference

AMRBench_QC modules — D2 D1

`git submodule @ 52001eb`
QUAST 5.3.0 · assembly statistics
CheckM2 1.1.0 · completeness, contamination
Bakta 1.12.0 (full DB) · gene prediction
Output: proteins (.faa) to CarveMe;
QC metrics to the summary (no filtering)

Solver check (once, first) — D6

2,500-binary MILP must solve optimally;
catches missing CPLEX or size-limited
editions before any genome is modelled
CPLEX threads capped at task CPUs

Orchestration — D8

Nextflow DSL2 on SLURM + Apptainer; one task per genome; -resume
Pinned versions; CPLEX image built locally from a licence
Retries with more memory; a genome that still fails is skipped
and listed in the summary (or --fail_fast)
Reference preparation runs once and is reused

proteins (.faa) + species

CarveMe 1.6.6 · carve (wrapped) — D7

DIAMOND 2.1.9 vs CarveMe BiGG DB
Reaction scores from BiGG GPRs
MILP carving of gramneg / grampos universe
--reference: harmonised species reference
--reference-score 0.0; no -g
Wrapper: MILP status recorded; error if
no model written (carve exits 0 on failure)
Output: draft model + DIAMOND table

Biomass replacement

Remove CarveMe 'Growth'
Add the reference biomass module:
biomass reaction, its lumping and
drain reactions, required sinks
(e.g. iYS854: 13 reactions)
Reference biomass = only objective
ATP maintenance (ATPM) bounds
from the reference

Reference-guided gap-filling — D3 D5 D9

CarveMe's own gapFill() MILP
Pool = universe + reference-only reactions
Cost = 1 / (1 + homology score + 1.0
if a reference reaction)
LB + biotin, uptake 100, growth 0.1 or more
Then medium stored with uptake 10
Added reactions listed with source,
score and candidate GPR

QC (COBRApy, report only)

SBML valid; objective correct
Growth equals build (reframed)
No growth without carbon
No ATP from nothing (EGC)
Mass/charge balance where
formulas/charges are set
Blocked reactions (FVA),
GPR coverage

Outputs

`<sample>.xml.gz`
`.gapfill.tsv`
`.build.json`
`.qc.json`
one summary table
for the run, incl.
failed genomes

harmonised reference · *biomass module, ATPM* · *pool, reaction sources, medium*

ONCE PER REFERENCE (6 MODELS; KpSC-pan SHARED BY KLEBSIELLA AND ENTEROBACTER)

Reference registry — D4

`assets/references.csv`
species: reference GEM file
biomass reaction ID
universe: gramneg / grampos
SHA-256 of the reference file
`assets/media.tsv: LB_btn`

Prepare reference

Check SHA-256; formulas and charges from SBML fbc fields; compartments x to C_x
Biomass module: biomass + lumping/drain reactions (alternative biomasses excluded)
Pool: universe + reference-only reactions; reference variants for ID clashes with
reference-specific chemistry; no sinks, no gene rules
Required sinks (greedy minimum) only if pool + module cannot grow
Stop the run unless reference and pool + module both grow on the medium

Prepared reference bundle

harmonised reference (SBML)
biomass module (SBML) + biomass/ATPM settings (JSON)
gap-fill pool (SBML) + origin of each pool reaction
validation: growth checks, required sinks, ID clashes,
balance diagnostics, software versions

DECISIONS (SETTLED 2026-10-22)

- D1 Proteins: Bakta 1.12.0, full database (AMRBench_QC modules)
- D2 QC: report CheckM2/QUAST and model QC only; no filtering
- D3 Medium: CarveMe LB + biotin for all species
- D4 References: iML1515, KpSC-pan v2 (+ Enterobacter), iCN718, iSD1509, iYS854, iDR478
- D5 Gap-fill cost: 1/(1 + homology + 1.0 for reference reactions)
- D6 Solver: CPLEX (SCIP accepts suboptimal solutions)
- D7 carve --reference-score: 0.0 (CarveMe default)
- D8 Compute: SLURM + Apptainer; cluster config supplied separately
- D9 Minimum growth for gap-filling: 0.1 (CarveMe default)

LEGEND

- Existing tool / pipeline
- resisto-GEM code (bin/)
- Data / inputs / outputs
- Dn Decision (see panel)
- Reference bundle used by a step

CarveMe 1.6.6 source: --reference changes only carving scores; -g gap-fills from the universe only; the biomass is the universe 'Growth'. Hence the orange steps.
Models are written without boundaryCondition species: with reframed's default output, COBRApy opens uptake of every extracellular metabolite (growth 50.2 instead of 3.48).
Details and verification: docs/method.md. Tool versions for AMRBench_QC are those of its modules at commit 52001eb.